

Quantum Bluff

FINAL PROJECT REPORT

University of Strasbourg

Bachelor of Science in Computer Science (L3)

Supervised project — reference specification: [Docs/Cahier_de_charge_6A_PI.pdf](#)

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Academic year 2025–2026

Document generated: 14 May 2026

Abstract

Quantum Bluff is a capstone web platform produced for the third-year Bachelor's degree in Computer Science (L3) at the University of Strasbourg. It combines an authoritative Texas Hold'em stack with extended casino-style modules (solo and multiplayer blackjack, slots, European roulette), structured tournaments with ancillary winner markets, social features (friends, invitations, peer loans), and progression systems (XP, daily challenges, leaderboards). This report explains how delivery evolved from an initial V-model framing to an agile-inspired workflow; which engineering tools preserved modularity and testability; and how runtime architecture separates the React client, Express REST endpoints, a Socket.IO gateway, Prisma-backed persistence, and Redis-backed coordination. Security controls (JWT and optional TOTP, segmented rate limits, idempotency on sensitive mutations, anti-cheat hooks) and an automated test harness are summarized alongside appendices that map each major domain to repository artefacts. The product uses **virtual chips** for pedagogical demonstration; it does not implement real-money gambling.

Résumé

Quantum Bluff est une plateforme web temps réel développée dans le cadre du projet de fin d'études de L3 informatique à l'Université de Strasbourg. Elle couvre le poker Texas Hold'em (tables cash, bots, paris latents), des modules type casino (blackjack solo et multijoueur, machine à sous, roulette européenne), des tournois structurés avec marchés annexes sur le vainqueur, des fonctions sociales (amis, invitations, prêts entre joueurs) et de la progression (XP, défis quotidiens, classements). Ce rapport présente l'évolution de la gestion de projet (phase en V initiale puis itérations inspirées du Scrum), les choix technologiques, l'architecture logicielle, la sécurité et les résultats des tests automatisés. Des annexes indexent les chemins du dépôt pour chaque sous-système. Le système repose sur une **monnaie virtuelle** à visée pédagogique et ne constitue pas un service de jeu d'argent réel.

Keywords: multiplayer games, Texas Hold'em, real-time web, WebSocket, TypeScript, React, Node.js, Express, Prisma, PostgreSQL, Redis, Socket.IO, software engineering, security, automated testing.

Mots-clés : jeux multijoueurs, Texas Hold'em, web temps réel, WebSocket, TypeScript, React, Node.js, Express, Prisma, PostgreSQL, Redis, Socket.IO, génie logiciel, sécurité, tests automatisés.

List of abbreviations

API	Application Programming Interface (HTTP/JSON in this project).
CI/CD	Continuous integration and continuous deployment (GitLab CI at repository root).
HUD	Heads-up display: in-table probability and hand-category feedback.
JWT	JSON Web Token used for HTTP and Socket.IO authentication.
ORM	Object–relational mapping (Prisma).
REST	Representational state transfer style for <code>/api/*</code> routes.
RTK	Redux Toolkit and RTK Query for client-side data fetching.
TOTP	Time-based one-time password (optional second factor).
TTL	Time-to-live for cached or in-memory sessions.
UI/UX	User interface and user experience layers in the React SPA.
2FA	Two-factor authentication (TOTP in production paths).

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1 Introduction

1.1 Context and Objectives

Quantum Bluff was developed as part of the final third-year Bachelor’s degree project (L3 Computer Science, French academic system) at the University of Strasbourg. The objective was to deliver a **real-time multiplayer gaming platform** with a professional-grade engineering posture: modular architecture, automated testing, observability hooks, security hardening, and collaborative delivery under time constraints.

The **initial product intent** (before scope expansion) was a Texas Hold’em poker platform: practice mode against AI with multiple difficulty levels, multiplayer cash tables (public/private), configurable blinds, minimum buy-in rules, a probability-oriented HUD, and a **hidden side-bet** subsystem. The technical goals included authoritative server-side rules, transactional chip movements, WebSocket-scale event delivery, and operational readiness (health checks, metrics, recovery paths).

Security and reliability were first-class constraints from the outset: JWT authentication, optional TOTP 2FA, anti-cheat and rate-limiting layers, idempotent handling for sensitive mutations, and schema validation on critical API paths.

1.2 Scope of this document

This report summarizes **management evolution, tooling, delivered capabilities, and architecture**. Exhaustive technical narratives (bots, probabilities, tournaments, hidden bets, etc.) are deferred to **Appendices A–M** to avoid repetition. Repository-relative file paths use `\path{...}` notation. The front matter includes an abstract (English and French), a list of abbreviations, the table of contents, and lists of figures and tables.

2 Methodology, traceability, and reading guide

2.1 Traceability to the initial specification

The baseline expectations for the Bachelor project are captured in the repository specification `Docs/Cahier_de_charge_6A_PI.pdf`. The delivered system **extends** that poker-centric charter: all original poker-facing commitments (cash tables, bots, HUD, hidden bets, real-time play) remain grounded in server authority, while additional modules (casino mini-games, tournaments, social lending, progression) were integrated under the same engineering constraints (tests, rate limits, transactional chip movements). Where the PDF and the code diverge, **the implementation and this report defer to the repository as the operational source of truth**, with internal notes under `Docs/detail_logique/` providing rationale.

2.2 Evidence hierarchy

Claims in the body are intentionally **outcome-oriented**. File paths anchor verifiable detail. Long-form auto-generated material (for example `Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md`) is treated as an *index*: useful for breadth, but not cited as automatically validated against every line of code.

2.3 How validation is reported

Automated suites and manual smoke scenarios are summarized in Section 10 and cross-checked against `Docs/RAPPORT_TESTS.md` and `Docs/rapport-validation.md`. Playwright end-to-end tests live under `client/e2e/`. CI behaviour is defined in `.gitlab-ci.yml`.

3 Project Evolution

3.1 Initial specifications (poker-first)

The early scope targeted: Texas Hold'em cash games; **AI practice** with four difficulty tiers (easy, medium, hard, expert) enforced in `server/src/routes/game.api.routes.ts` and `server/src/routes/bot.routes.ts`; public/private multiplayer rooms with SB/BB configuration and minimum balance rules; **real-time** updates via Socket.IO; a **Quantum HUD** for win-equity and hand-category feedback (`client/src/components/QuantumHUD.tsx`); and **hidden bets** resolved server-side (`server/src/poker/hiddenBets/`).

3.2 Scope expansion (casino, tournaments, social, progression)

The delivered system adds: **Blackjack** solo and **multi-table** lobbies (`client/src/pages/Blackjack.tsx`, `client/src/pages/BlackjackMultiLobby.tsx`, `client/src/pages/BlackjackMultiTable.tsx`); **slot** and **roulette** flows reachable from the mini-games hub (`client/src/pages/MiniGames.tsx`); **tournaments** with waiting room, in-tournament table, results, and **winner prediction side bets** (`client/src/features/tournament/`); **leaderboards**; **XP**, badges, daily challenges, daily login streak; **friends**, invitations, join requests, **peer loans** with interest/repayment hooks; **profile** editing with server-stored avatars; **2FA TOTP**; **feedback** and **player reports**; **wallet ledger**, gift codes, and dev recharge helpers; **i18n** in five locales (`client/src/i18n/locales/`); **tutorial** flow (`client/src/pages/TutorialGame.tsx`, `client/src/features/tutorial/`); **admin web console** and packaged targets (**Electron**, **Capacitor**) per `client/package.json`.

3.3 Why the narrative moved from “catalog” to “appendices”

The codebase grew across **client**, **server**, **database**, and **docs/** technical notes. Listing every subsystem in prose would duplicate `docs/ARCHITECTURE.md` and the structured logic notes under `Docs/detail_logique/`. The body therefore stays outcome-oriented; appendices map each major subsystem to code anchors.

4 Project Management

4.1 Initial V-model phase

The project began with a **V-model style** framing: business requirement gathering with end users in mind, benchmarking comparable products, brainstorming sessions **validated collectively** by the team, and a written requirements package plus an initial Gantt chart. A **technical lead** was appointed to structure the MOE (implementation) track while the project manager covered the MOA (needs, prioritization, specification) track until the specification baseline stabilized.

4.2 Transition to an agile-inspired workflow (Jira, Scrum practices)

After an intensive early phase (approximately **two months** of V-model-oriented execution on a **four-month** calendar), the team concluded that strict phase separation slowed integration of new gameplay and infrastructure ideas. The organization shifted toward an **iterative, Scrum-inspired workflow**: **Jira** for epics, user stories, and time-boxed sprints; backlog ordering by **value vs. risk**; twice-weekly synchronization meetings for decisions, unblockers, and demo-ready increments. The project manager studied agile practice material (including *Scrum Life*) to align rituals with the team's reality rather than copying a textbook process wholesale.

4.3 Planning, coordination, and reporting

Coordination relied on: recurring meetings (~twice per week), shared technical documentation under Docs/, internal validation sessions, and continuous integration feedback via `.gitlab-ci.yml` at the repository root. Work was distributed dynamically across frontend, backend, and infrastructure tasks because multiplayer features require tight coupling between UI state, socket events, and persistence.

4.4 Product management mechanics

Backlog prioritization balanced user-visible value (new game modes, social features) against enabling work (Redis adapter, rate limits, recovery jobs, observability). Cross-cutting concerns (security, idempotency, anti-cheat) were escalated early to avoid late integration risk.

5 Development Tools and Technologies

5.1 Frontend and backend

Client: React 19, TypeScript, Vite 7, Tailwind CSS, Radix UI primitives (`client/src/components/ui/`), react-router-dom v7, Redux Toolkit with RTK Query (`client/src/services/api.ts`), socket.io-client (`client/src/services/socket.ts`), i18next (`client/src/i18n/config.ts`).

Server: Node.js (ESM), Express 4, Socket.IO 4, Prisma 6 with PostgreSQL (`server/prisma/schema.prisma`), Redis clients for rate limiting, action logging, and the Socket.IO Redis adapter (`server/src/index.ts`), Zod validation, JWT auth, bcrypt password hashing, sanitize-html where applicable.

5.2 Database and infrastructure

Local PostgreSQL via `database/docker-compose.yml`. Redis for multi-instance coordination and abuse controls. Optional OpenTelemetry and Prometheus-oriented metrics endpoints (`server/src/observability/`).

5.3 Version control and collaboration

Git repository with GitLab CI configuration at `.gitlab-ci.yml`. Beyond unit tests and security scans, the `docker-build` stage uses **Kaniko** (`gcr.io/kaniko-project/executor:v1.23.2-debug`) to build and push OCI images **without a Docker daemon** on GitLab runners: jobs `docker:build` (server), `docker:build:client` (Vite frontend with `VITE_API_URL` build-arg), and `docker:build:ai-service` publish to the project's GitLab Container Registry for VM pull-based deployment. Documentation and logic notes live under `Docs/` (including `docs/ARCHITECTURE.md` and `Docs/detail_logique/`).

5.4 Testing and deployment

Client: Vitest (`client/vite.config.ts` coverage over `src/**/*.ts`), Playwright (`client/e2e/`). Server: Jest with Supertest integration suites (`server/src/__tests__/`). Desktop/mobile packaging: **Electron** (electron-builder) and **Capacitor** 8 for native shells; scripts in `client/package.json` produce **Windows installer** (`.exe NSIS`), **macOS disk image** (`.dmg`) and companion artefacts, and **Linux** targets (AppImage, `.deb`); Capacitor opens **iOS** (xcodeproj) and **Android** (gradle) projects after `npm run build:cap` and `cap sync`. See Section 9 and Appendix L.

6 Use of Artificial Intelligence

6.1 Programmed in-game poker AI (bots)

The product ships **four deterministic difficulty tiers**—easy, medium, hard, expert—implemented entirely in TypeScript (`server/src/logic/botAI.ts`) and exposed through `server/src/routes/bot.routes.ts`. Each decision receives the visible hole cards, community cards, pot geometry (`potSize`, `callAmount`, `minRaise`), stacks, seat index, and player count. A shared primitive maps any partial board to a scalar strength by normalising the evaluator score against the theoretical maximum ($S_{\max}$ in Appendix C). Easy and medium tiers inject tunable randomness (loose calls, naive bluffs); hard tightens fold thresholds; **expert** adds deeper pot-odds heuristics before any optional remote call.

6.2 Expert tier: Python oracle and local oracle

`server/src/services/botAi.service.ts` wraps `decideBotAction`. For **expert**, the service first tries an **optional HTTP micro-service** when `AI_SERVICE_URL` is set: it POSTs a compact JSON payload (holes, board, stacks, street, etc.) to `/predict/poker`, validates the JSON response with **Zod** (`action`, `amount`, `confidence`, `style`, `reason`), converts `ALL_IN` into legal raise/call/fold semantics, and logs latency. If the URL is unset, the HTTP call fails, or validation fails, the server **falls back** to the pure-TypeScript `expertBotDecision`.

When hole cards of opponents are known (practice/oracle contexts), `expertOracleDecision` compares the hero strength to a **composite opponent strength** built from sorted opponent strengths $\{s_1 \geq \dots \geq s_n\}$: for $n \geq 3$ it blends the best and median components with a player-count-dependent weight w (Appendix C), then applies pot-odds and pressure ratios to choose value raises, bluffs, calls, or folds. This path avoids the “always fold” failure mode of heuristics that ignore revealed cards.

6.3 Expert neural policy (PyTorch MLP, optional service)

When `AI_SERVICE_URL` points to the FastAPI bundle under `server/ai-service/`, the expert path can run a **trained multilayer perceptron** (`ExpertPolicy` in `server/ai-service/poker/train.py`, weights serialised to `server/ai-service/model/expert_bot.pt`). The vectoriser `features.py` exports `FEATURE_NAMES`: **36** numeric descriptors (streets, equity, pot odds, SPR, board texture, action frequencies, etc.). The network stacks **four linear layers** separated by ReLU: $36 \rightarrow 96 \rightarrow 64 \rightarrow 32 \rightarrow 4$ logits. Counting post-input activation units (hidden + output heads) gives $96 + 64 + 32 + 4 = 196$ **neurons**. Dropout ($p=0.06$ then 0.04) is applied after the first two hidden blocks during training only. The four outputs correspond to the internal action indices distilled from the expert teacher (`expert_rules.py`); inference-time mixing with heuristics and temperature sampling is described in `server/ai-service/poker/decision.py`. The compact trainable weight count in the linear stack is **11 972** scalar parameters (weights plus biases). Figures 1 and 2 show **flowcharts**: tensor pipeline inside the expert policy, then the Node $\leftrightarrow$ Python decision path with branching. The

compact trainable weight count in the linear stack is **11 972** scalar parameters (weights plus biases).

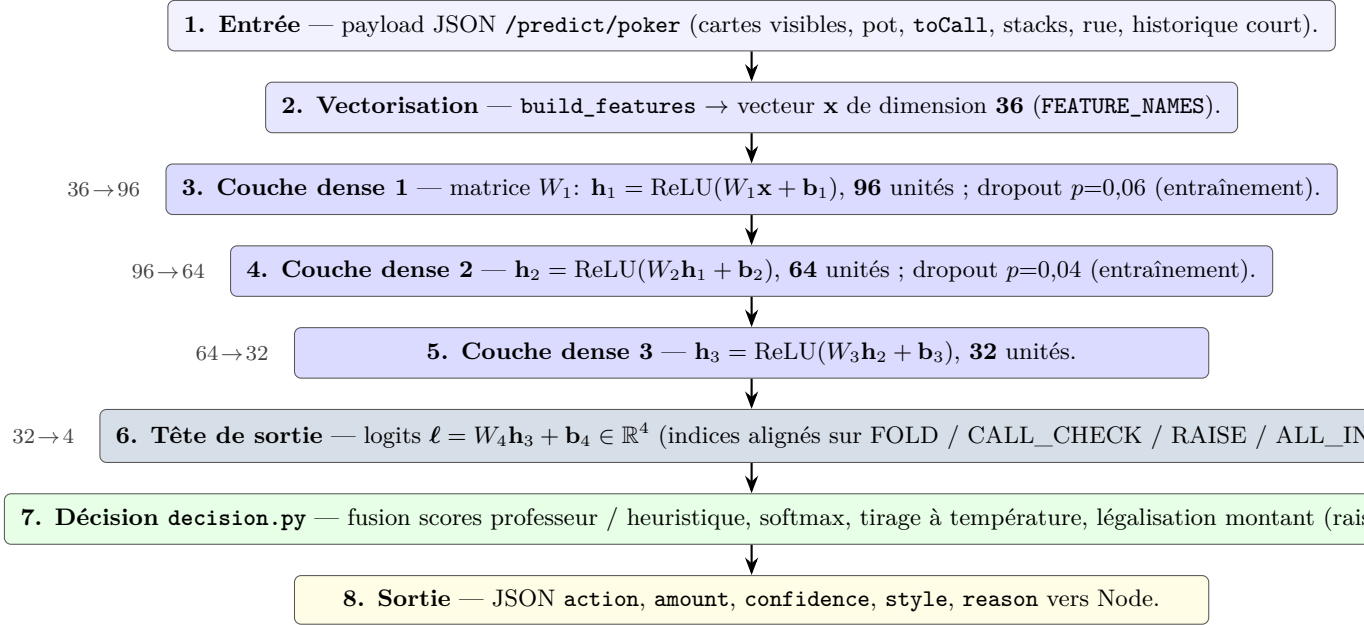


Figure 1: Flux de données à travers la politique MLP expert (**ExpertPolicy**): 4 transformations linéaires successives, **196** neurones après la couche d’entrée (96+64+32+4), puis post-traitement métier avant réponse HTTP.

6.4 AI-assisted engineering (tooling)

Team members used modern **coding assistants** (e.g., IDE-integrated LLMs) for drafting, refactors, and documentation. Generated output was treated as **untrusted**: reviewed in merge requests, validated by automated tests, and rejected when it contradicted server authority or security invariants.

6.5 Limits and verification

The **easy/medium/hard** tiers and the TypeScript **expert** fallback contain **no trainable neurons**: they are hand-crafted policies. Only the optional Python bundle instantiates the MLP in Figure 1. Regardless of path, bot traffic is **rate-limited**, responses pass through **Zod and sanitisation** before touching authoritative state, and assistant-generated repository edits were never merged without human review and automated tests.

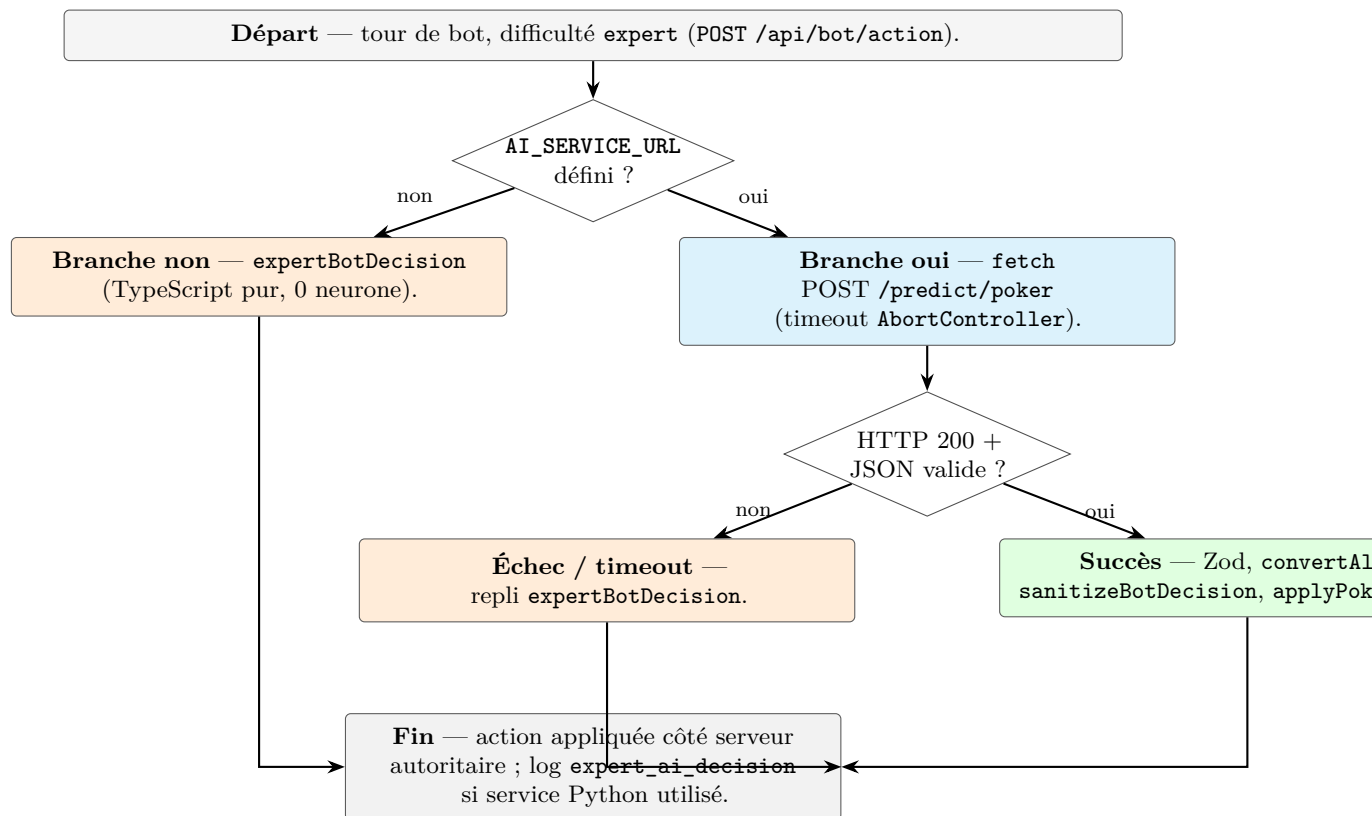


Figure 2: Diagramme de flux décisionnel côté Node pour le niveau `expert` (orchestration `botAi.service.ts`, repli heuristique, durcissement).

7 Platform Functionalities

Table 1 lists major domains. Details appear only once, in the referenced appendix, to prevent duplication. **Appendices A–N** also record implemented formulas: closed-form **slot RTP** and unified **roulette RTP**, Monte Carlo equity with an error bound for the **Quantum HUD**, **Quantum Combi** exact probability plus its place in the hand-ranking cascade, **tournament bracket** sizing and RNG, **code coverage** definitions, and the source files where each is coded.

Table 1: High-level functional inventory (representative surfaces).

Domain	User-visible capability	Primary UI entry	Deep dive
Poker cash	Multiplayer tables, pot logic, chat, actions	<code>/game,</code> <code>/waiting-room</code>	Appendix B
Poker bots	Practice tables, four difficulties	<code>/bot-configuration</code>	Appendix C
HUD	Win chance and hand category estimates	In-game HUD	Appendix D
Hidden bets	Side markets, pricing, settlement	In-game panel, <code>/results</code>	Appendix E
Blackjack	Solo + multi lobby/table	<code>/blackjack,</code> <code>/blackjack/</code> lobby, <code>/blackjack/</code> <code>table/:gameId</code>	Appendix G
Casino	Slot and roulette	<code>/minigames</code>	Appendix H
Tournaments	Waiting, play, results, winner side bets	<code>/tournaments/..</code> .	Appendix F
Social	Friends, invites, loans	<code>/friends,</code> lobby flows	Appendix I
Progression	XP, badges, daily challenges, login streak, leaderboard	<code>/profile,</code> <code>/leaderboard</code>	Appendix J
Account	Auth, 2FA, avatars, profile edit	<code>/auth,</code> <code>/edit-profile</code>	Appendix K
Admin	Operator web console (JWT admin role)	<code>/admin/console</code>	Appendix L
Ops	Metrics, health, CI, packaging	N/A (infra)	Appendix L
Other UX	Gift codes, free recharge, post-game rating modal	<code>client/src/</code> <code>components/</code> <code>Layout.tsx,</code> lobby	Appendix L

Domain	User-visible capability	Primary UI entry	Deep dive
Feedback	User feedback and player reports	Lobby/settings flows	Appendix L
Tournament UX	ZipRush mini-game while waiting	<code>client/src/features/tournament/components/ZipRushMiniGame.tsx</code>	Appendix F
Demo	Example and deal teaching routes	<code>/game-example</code> , <code>/game-deal</code>	Appendix B
A11y	Accessibility, table theming, ambient audio	<code>client/src/contexts/AccessibilityContext.tsx</code> , <code>TableThemeContext.tsx</code> , <code>MusicContext.tsx</code>	Appendix A

8 System Architecture

8.1 Monorepo layout

The repository splits into `client/` (Vite SPA), `server/` (Express + Socket.IO gateway), and `database/` (local Postgres). Client bootstraps providers in `client/src/main.tsx` and routes in `client/src/App.tsx`. Server composes middlewares and REST mounts in `server/src/index.ts`, then attaches `server/src/sockets/game.gateway.ts` to the HTTP server.

8.2 Structured tournaments (outline)

Multi-table Texas Hold'em tournaments are **first-class server domain**: Prisma-backed registration and fees (`server/src/tournament/tournament.service.ts`), a periodic **scheduler** that auto-starts due events (`server/src/tournament/tournament.scheduler.ts`), runtime orchestration that builds bracket rounds and assigns `GameTable` instances (`server/src/tournament/tournament.runtime.service.ts`), deterministic **table splitting** and re-seating (`server/src/tournament/bracket/`), winner-bet settlement (`server/src/tournament/winnerBets/`), and boot-time **recovery** (`server/src/tournament/tournament.recovery.service.ts`). The client module `client/src/features/tournament/` binds waiting room, in-tournament play, results, and ZipRush. Exact bracket arithmetic, RNG seeding, and coverage metrics are in Appendices F and M.

8.3 Real-time communication

Real-time communication is implemented with Socket.IO on the `/socket.io` path. The design follows an **authoritative server** model: the browser can request an action, but the server decides whether the user is authenticated, whether the action is legal, how chips move, and which sanitized state each participant is allowed to see. This is essential for a poker platform, because private cards, turn order, pot resolution and hidden side bets must never be trusted to the client.

The socket handshake uses the same identity model as the REST API. A player token is sent either through `handshake.auth.token` or through an `Authorization: Bearer ...` header, then validated by `server/src/middleware/socketAuth.middleware.ts` and the gateway-level checks in `server/src/sockets/game.gateway.ts`. The server verifies token revocation, JWT issuer/audience/type, role separation (admin tokens are rejected on player sockets), user existence, and active suspension status before attaching the authenticated user id to the socket. Once connected, each user joins a private room `user:{userId}` for targeted events such as invitations, tournament assignments, warnings, and social notifications.

Game state synchronization is organized around Socket.IO rooms. A poker or blackjack table has its own room; lobby and tournament features use dedicated rooms or user-targeted emissions. On join and reconnect, the gateway sends a full sanitized snapshot instead of relying only on incremental events. For poker, active runtime controllers are stored through `server/src/shared/activeGames.ts`, while serializable table snapshots are pushed to `server/src/shared/pokerStateStore.ts`. For blackjack, recovery and snapshot logic is implemented

under `server/src/blackjack/recovery/` and `server/src/blackjack/services/`. This makes the client resilient to refreshes, temporary disconnections and process-level recovery paths.

Synchronization also protects confidentiality. The server computes a different snapshot depending on the receiving socket: a seated player receives their own private hand, while spectators or other players receive a sanitized view. Broadcast helpers such as `server/src/sockets/gameIo.registry.ts` and the gateway methods emit both legacy and current event names where needed (`GAME_UPDATE`, `GAME_STATE_UPDATED`) so older UI flows remain compatible during refactors. Redis support extends Socket.IO to multi-instance deployments through `server/src/config/socketIoRedis.ts`; when Redis is unavailable, the application keeps a local fallback for development and degraded operation.

8.4 Persistence and integrity

Prisma models capture users, games, tournaments, loans, hidden bet tickets, wallet ledger rows, and more (see `server/prisma/schema.prisma`). Critical monetary flows favor server-side orchestration and short transactions.

8.5 Data model overview (Prisma)

The persistence layer is relational PostgreSQL accessed through Prisma. Representative entities include: **User** (credentials, profile, chips, XP fields, avatar storage keys), **Game/seat** models for poker tables, **BlackjackRoom** and related seat/invitation models for multiplayer blackjack, tournament tables for bracket state and winner-bet tickets, **Friendship** and messaging hooks, loan and ledger tables for social/casino flows, and audit-friendly structures for wallet history. Full cardinality and migration history are only authoritative in `server/prisma/schema.prisma` and the migration folder.

8.6 Diagram

Figure 3 shows a **runtime flow diagram** (logical path, not deployment topology). For more detail, see `docs/ARCHITECTURE.md`.

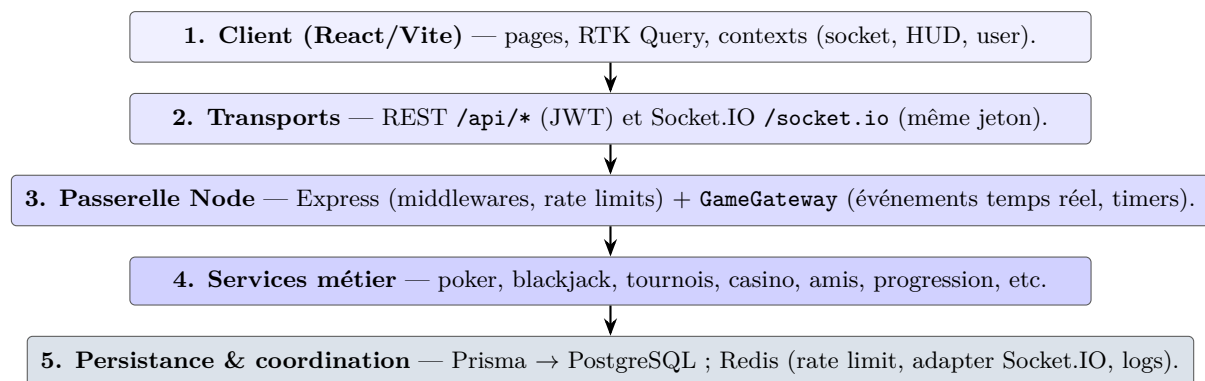


Figure 3: Flux d'exécution monorepo : interface → API / WebSocket → passerelle et services → données.

9 Deployment and operational runbook

9.1 Local development stack

The root script `npm run dev` (see `package.json`) launches PostgreSQL through `database/docker-compose.yml`, a lightweight helper script (`scripts/dev-ai.js`), the Node/Express server (`server/`), and the Vite client (`client/`) under concurrently. Redis should be available for rate limiting, action logging, and the Socket.IO Redis adapter when exercising multi-instance paths (`server/src/config/socketIoRedis.ts`). First-time setup, tester onboarding, and troubleshooting are documented in `README.md`, `Docs/SETUP_TESTEUR.md`, and `Docs/QUICK_TEST_GUIDE.md`.

9.2 Production and desktop delivery

VM-oriented deployment (Node, PostgreSQL, Redis, PM2), environment files, and Electron auto-update mechanics are described in `Docs/DEPLOY.md`. The pipeline builds runnable **container images** on `develop`, `main`, and `release/*` using **Kaniko**: each job runs `/kaniko/executor` with `-context` pointing to `server/`, `client/`, or (on `release/*` only) the Python AI bundle under `server/ai-service/`, authenticates to the GitLab registry via `/kaniko/.docker/config.json`, and pushes immutable SHA-tagged images plus rolling `:latest`, `:client-latest`, or `:ai-latest` aliases so the VM can `docker pull` without rebuilding from source on the host. Reverse-proxy samples appear under `nginx/` when applicable.

9.3 Electron and Capacitor artefacts

The same React UI is built under three Vite modes: default web, `-mode electron`, and `-mode capacitor`. **Electron** (`client/electron.cjs`, `electron-builder` in `client/package.json`) targets **Windows** (NSIS → `.exe` installer), **macOS** (`.dmg` / signed artefacts per builder defaults), and **Linux** (AppImage and `.deb` as configured under `build.linux.target`). Published updates are served from the generic URL declared in `build.publish` for auto-update (`electron-updater`). **Capacitor** wraps the static build for **iOS** and **Android**: `npm run cap:sync` copies web assets into native projects; `cap:ios` / `cap:android` open Xcode and Android Studio. Repository scripts `scripts/package-ios-rendu.sh` and `scripts/package-android-rendu.sh` support course delivery packaging. Appendix L lists the npm entry points.

10 Security, Testing and Stability

10.1 Authentication and anti-cheat

Authentication is centralized around short-lived JWT access tokens generated by `server/src/auth/jwt.service.ts`. Tokens use HS256 with configured issuer and audience, a mandatory `type: access`, and a subject aligned with the user id. HTTP routes only accept the standard `Authorization: Bearer ...` format; malformed or raw tokens are rejected early by `extractBearerToken`. The protected-route middleware `server/src/middleware/auth.middleware.ts` then checks the token blacklist, verifies the JWT, rejects admin-role tokens on player routes, confirms that the user still exists in PostgreSQL, and denies suspended accounts with an explicit `ACCOUNT_SUSPENDED` response.

The account layer combines several controls rather than relying on a single mechanism. Passwords are hashed with `bcrypt` in `server/src/routes/auth.routes.ts`; registration, password reset and profile password changes use Zod validation from `server/src/validation/auth.validation.ts`, including minimum length, uppercase, numeric and special-character requirements. Optional TOTP two-factor authentication is implemented in `server/src/routes/twofa.routes.ts`; pending setup secrets are stored temporarily in Redis, and disabling 2FA requires both the current password and a valid one-time code. Logout and token revocation use `server/src/auth/tokenBlacklist.ts`, which hashes tokens before storing blacklist entries in Redis with an in-memory fallback for degraded conditions.

The same identity policy applies to real-time traffic. `server/src/middleware/socketAuth.middleware.ts` rejects revoked tokens, invalid JWTs, admin tokens, missing users and suspended users before a socket can interact with game rooms. In `server/src/sockets/game.gateway.ts`, player-sensitive events compare `socket.userId` with the submitted `playerId`; mismatches are logged as suspicious actions and refused. This prevents a modified client from joining or acting as another player.

Anti-cheat is implemented as a layered abuse-detection system. The HTTP middleware `server/src/middleware/antiCheat.middleware.ts` reads the authenticated user when available, blocks active bans, records the last observed IP address, and delegates multi-account detection to `server/src/services/antiCheat.service.ts`. The service raises alerts when several distinct accounts appear on the same non-local IP and when player reactions are unrealistically fast. Repeated alerts trigger a temporary suspension through the `bannedUntil` field. At the game gateway level, `server/src/utils/antiCheat.ts` tracks action bursts (more than eight actions in three seconds by default), while the poker action handler refuses excessive actions and logs the incident.

Abuse controls are completed by operational safeguards: Helmet and CORS allowlists in `server/src/index.ts`; segmented rate limits for authentication, bots, slot, roulette, blackjack, hidden bets and multiplayer blackjack; idempotency middleware for sensitive mutations (`server/src/middleware/idempotency.middleware.ts`); and an admin surface that exposes cheater/report information without reusing player tokens. These measures do not replace a professional penetration test, but they provide a coherent defence-in-depth baseline for the academic prototype.

10.2 Testing and debugging

Broad automated coverage on server and client; Playwright smoke tests under `client/e2e/`. Additional manual scenarios are documented under `Docs/POKER_SCENARIOS.md` and related files.

10.3 Automated test inventory and CI (reference snapshot)

Table 2 reflects a **May 2026** run (`cd server && npm test, cd client && npx vitest run`). Historical line counts also appear in `Docs/RAPPORT_TESTS.md`; refresh that file after large suite changes.

Layer	Runner	Test files / suites	Tests passed
Backend (<code>server/</code>)	Jest	51	520
Frontend (<code>client/</code>)	Vitest	13	83
Total unit/integration	—	64	603

Table 2: Automated unit and integration test inventory (May 2026 snapshot).

10.4 Code coverage policy (server)

Jest collects coverage only on **logic-heavy paths** listed in `server/jest.config.cjs` (`collectCoverageFrom` poker/tournament bracket utilities, validation, selected `src/`utils, etc.; some monolithic controllers are excluded so the metric tracks maintainable units). **Global thresholds** require at least 80% lines, statements, and functions, and 65% branches, or CI fails. Table 3 shows the latest measured aggregate on that scope. Appendix M defines the coverage ratios and relates them to tournament bracket tests.

Layer	Lines	Statements	Functions	Branches
Server (Jest scoped paths)	86.56%	83.69%	88.75%	67.90%
Required minimum (threshold)	80.00%	80.00%	80.00%	65.00%

Table 3: Server Jest coverage aggregate on `collectCoverageFrom` (May 2026; `npm run test:coverage` in `server/`).

Client: running `vitest run -coverage` over the whole `src/` tree yields a **low global percentage** (~4% lines in a May 2026 run) because most UI and socket flows are covered by Playwright, manual poker scenarios, and integration with the live API rather than by isolated Vitest files. Per-file reports under `client/coverage/` still highlight modules with strong unit tests (e.g. `iban`, `avatars`).

GitLab CI configuration at `.gitlab-ci.yml` encodes build and test gates for merge requests; the `docker-build` stage uses **Kaniko** for reproducible, rootless image builds pushed to the GitLab Container Registry. A consolidated validation narrative for bot flows and UI smoke checks appears in `Docs/rapport-validation.md`.

10.5 Performance and stability

Redis-backed Socket.IO adapter, structured logging, Prometheus metrics endpoint, readiness endpoints, graceful shutdown hooks (`server/src/index.ts`).

11 Technical Challenges

Concurrency around shared tables: poker and blackjack runtimes use locking strategies (see `server/src/poker/services/pokerTableLock.service.ts` and blackjack lock services) to prevent split-brain chip mutations.

Recovery after process restarts: blackjack and tournament subsystems implement recovery services invoked at boot (`server/src/index.ts`).

Hidden bet settlement coupling: market resolution must align with poker street transitions; implemented in hidden-bet resolver modules (Appendix E).

Internationalization without clobbering user preference: guest screens (StartScreen, Auth, global loader) force English using `react-i18next` with a fixed `lng: 'en'` override so the global language stored for authenticated use is unchanged (`client/src/pages/Auth.tsx`, `client/src/contexts/LoaderContext.tsx`).

11.1 Implemented solutions

The project was completed by decomposing the platform into domain modules while keeping one consistent runtime architecture. The React client is responsible for navigation, presentation, forms, table interaction and user feedback; the Express API exposes stable REST endpoints; Socket.IO carries live events; Prisma/PostgreSQL stores durable state; and Redis is used where coordination, rate limiting, token revocation or distributed socket delivery are needed. This avoids a single monolithic “game page” and lets poker, blackjack, casino mini-games, tournaments, social features and progression evolve without losing a shared technical foundation.

For poker, the implemented solution was to keep the table authoritative on the server. Game clients render state and collect intentions; they do not compute final pot distribution, determine winners, grant chips, validate hidden bets, or decide whether a player may act. Poker actions are routed through `server/src/poker/services/pokerActionOrchestrator.service.ts` and the gateway, while table controllers under `server/src/logic/` keep the authoritative model. Hidden bets were implemented as a separate server-side subsystem under `server/src/poker/hiddenBets/`, so side markets can be priced, placed and resolved without leaking private poker information to the browser.

For synchronization issues, the implemented solution was a combination of room-based events, full snapshots, and recovery stores. Every important table update is broadcast to the relevant Socket.IO room, but reconnecting clients also receive a complete sanitized snapshot. Poker runtime snapshots are serialized through `server/src/poker/services/pokerStateSync.service.ts` and stored behind `server/src/shared/pokerStateStore.ts`; blackjack has its own snapshot repository and boot-time recovery service. This reduces the risk of clients rebuilding state from stale partial events.

For the casino extension, each game was isolated behind its own route and service layer instead of being mixed into the poker engine. Slot and roulette use dedicated logic modules and API routes, with server-side payout calculation, stake caps and wallet updates. Blackjack exists in both solo and multiplayer forms: solo blackjack is a direct server-controlled game

loop, while multiplayer blackjack uses table rooms, seats, snapshots and recovery services under `server/src/blackjack/`. The result is that all casino features share the same wallet and abuse-control principles while still keeping their own rules and UI flows.

For tournaments, the solution was to separate registration, waiting state, table assignment, runtime progression, finalization and rewards. Tournament screens live under `client/src/features/tournament/`; server-side orchestration is handled by `server/src/tournament/`. Socket.IO rooms such as `tournament:{id}` and `user:{userId}` are used to notify players about roster updates, table assignment, next rounds, cancellation and completion. Winner side bets were implemented as an additional tournament service rather than as ad hoc UI state, which keeps settlement tied to the real tournament result.

For social and lobby features, the project added friends, invitations, join requests, blocking/reporting logic, presence/activity, private warnings around blocked users, and peer loans. The solution combines REST routes for durable operations with socket notifications for live UX: a user can receive invitations, friend status changes, lobby activity updates and warning prompts without refreshing the page. Reports and moderation data are persisted so they can be reviewed from the admin surface rather than disappearing as client-only events.

For progression and retention, XP, badges, daily challenges, daily login streaks, leaderboards, gift codes and free recharge flows were implemented as separate modules. This keeps the virtual economy visible to the user while still routing rewards through server-side validation and persistence. The client pages display progress and rankings, but the server remains responsible for caps, eligibility and balance-changing side effects.

For security and anti-cheat, the final solution is deliberately redundant: HTTP middleware, socket middleware, gateway checks and domain services all re-check the assumptions that matter in their own layer. A forged REST body cannot replace the JWT identity when creating a waiting room; a forged socket payload cannot act for another player; a suspended account is rejected at login, REST middleware and socket connection time; and suspicious behavior is persisted as anti-cheat alerts for admin review or automatic temporary suspension.

For monetary integrity, chip-changing operations are handled server-side and backed by persistence patterns such as Prisma transactions, wallet ledger entries, idempotency keys and domain-specific settlement services. This pattern is visible in casino wallet modules, hidden-bet placement/resolution, tournament rewards and social loan flows. The common rule is that client-side balance displays are treated as cached UI, while the database and server-side services remain the source of truth.

For the interface, the solution was to keep the first screen after login as a functional lobby rather than a marketing page. The lobby aggregates poker rooms, blackjack rooms, tournaments, friends, invitations and daily challenges; deeper pages remain dedicated to gameplay, profile, friends, leaderboard and admin operations. Internationalization is handled through `i18next` resources, and reusable UI components reduce duplication across poker, casino and social flows.

Finally, the project added validation and test coverage around these solutions. Schema tests protect authentication validation, integration tests cover waiting-room identity trust, and real-time gateway tests verify socket behavior with authenticated users and runtime snapshots.

The remaining improvements would be a formal threat model and external penetration test before public deployment, broader end-to-end tests for the full lobby-to-game journey, and a polished observability dashboard for production operations; this is estimated at approximately two to three additional weeks depending on the desired depth.

12 Ethics, privacy, and academic integrity

12.1 Virtual economy and regulatory framing

Quantum Bluff uses **in-game virtual chips** for learning and demonstration. It is not a licensed real-money gambling operator; features that resemble casino products are implemented in an **academic engineering** context and should be evaluated as software artefacts, not as a commercial gaming offer.

12.2 Personal data

Accounts store identifiers needed for authentication and gameplay (email, username, hashed password, optional TOTP secrets, profile metadata). Operators deploying a public instance must align hosting, retention, and breach notification with applicable law (including GDPR for EU users). This report does not replace a data protection impact assessment.

12.3 Integrity of authorship and AI assistance

Section 6 states how large language models were used as drafting assistants. All merged code remained subject to human review, tests, and repository governance. Each author is expected to sign off on their contribution statement in Section 13.

13 Individual Contributions

13.1 Omar – project management and architecture alignment

Organizational narrative: Section 4. Beyond ceremonies and backlog arbitration, Omar coordinated **cross-cutting technical commitments**: early adoption of Redis for Socket.IO horizontal scale-out and rate limiting, alignment between Prisma schema changes and gameplay features, and risk reviews before widening scope (tournaments, casino mini-games, loans). Interface with the appointed technical lead focused on sequencing risky merges (socket gateway, recovery jobs) away from demo deadlines.

13.2 Yigit – Debugging and Quality Assurance

Yigit mainly contributed to debugging, testing and quality assurance throughout the development of Quantum Bluff. A large part of his work focused on identifying gameplay issues, reproducing bugs and validating the stability of the platform after new implementations. He participated actively in testing multiplayer poker games, tournaments and mobile responsive interfaces in order to ensure that the application remained functional and stable in different gameplay situations.

One of his main contributions concerned tournament synchronization and hand transition debugging. Several issues appeared during development, especially after showdown phases, during automatic hand restarts and final table transitions. Yigit worked on testing different edge cases, reproducing synchronization problems and validating the implemented fixes. Additional tests were performed to ensure that tournaments could continue correctly without requiring page refreshes and that action buttons remained functional after each hand.

He also contributed significantly to responsive design improvements on mobile devices. Different interface problems were identified during testing, including overlapping buttons, unreadable token displays and spacing issues between the poker table and gameplay controls. Yigit helped optimize the layout for smaller screens while preserving the overall user experience and keeping the desktop version stable.

In addition, he participated in Git integration, merge request validation and final testing sessions before deployment. His work helped improve the overall reliability, usability and quality of the final version of Quantum Bluff.

13.3 Other contributors

The repository does not contain signed-off, per-person effort logs, story point assignments, or validated textual attributions beyond the narratives above for Omar and Yigit. To satisfy examination requirements, the table below must still be **completed and countersigned by the team** for remaining members (replace placeholder cells).

Member	Primary areas (self-declared)	Verifiable artefacts (commits, MRs, modules)
Linda	<i>[to complete]</i>	<i>[to complete]</i>
Massi	<i>[to complete]</i>	<i>[to complete]</i>
Abdoulaye	<i>[to complete]</i>	<i>[to complete]</i>
Soheil	<i>[to complete]</i>	<i>[to complete]</i>
Mohamed	<i>[to complete]</i>	<i>[to complete]</i>
Azra	<i>[to complete]</i>	<i>[to complete]</i>
Yigit	Debugging, QA, tournaments, responsive UI, Git/MR reviews	See narrative above; playtests on <code>client/src/pages/Game.tsx</code> , <code>client/src/features/tournament/</code> , deployment regressions

Table 4: Individual contributions: expanded statements for Omar and Yigit appear as subsections above; other rows remain self-declared placeholders until completed.

14 Limitations and Future Work

14.1 Current limitations

- **Documentation volume vs. examiners' time:** internal generated reports (e.g., `Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md`) are useful as indexes but are not all human-curated line-by-line.
- **Dual loader providers:** `client/src/main.tsx` and `client/src/App.tsx` both wrap `LoaderProvider`; harmless but confusing—should be deduplicated.
- **Packaging surface area:** Electron/Capacitor increase release engineering work (signing, store policies) beyond the core web path.

14.2 Future improvements

Hardening continuous deployment secrets, expanding observability dashboards, deeper bot modeling (still code-first), and richer tournament analytics.

15 Conclusion

Quantum Bluff exceeded its initial poker-only charter by integrating multiple casino modes, tournaments, social lending, progression systems, and operational hardening while keeping a monorepo with clear separation between UI, API, and authoritative game logic. The project demonstrates that a student team can approach industry-like constraints—real-time state, monetary integrity, security, and observability—when management explicitly trades early waterfall rigidity for controlled agile iteration.

Lessons retained for similar efforts: (1) anchor scope changes to **recoverable increments** (feature flags, migrations, recovery jobs); (2) treat generated documentation as **hypotheses** until cross-read against code; (3) keep **security and idempotency** on the critical path rather than as end-of-project polish.

Acknowledgements

The team thanks **Jason Golda** for academic supervision of this L3 capstone, the University of Strasbourg Computer Science faculty for the project framework, the open-source maintainers of the libraries listed in Section 5, and peers who provided feedback during playtests. Replace or extend this paragraph with institution-specific acknowledgements as required by the course booklet.

A Repository map and routing

A.1 Programmed routing and shell

Client routes are declared in `client/src/App.tsx`: public splash and auth (`/`, `/auth`, `/auth/admin`); authenticated gameplay and hub routes under `ProtectedRoute` with `Layout` (lobby, poker, blackjack, tournaments, mini-games, profile, friends, admin console). Admin routes use `AdminProtectedRoute`. Global providers stack in `client/src/main.tsx` (error boundary, loader, Redux, toasts, user, socket, Quantum HUD, hidden bets, audio) with an additional nested loader/invitation shell in `App.tsx`. Server REST mounts, rate-limit tiers, and health/metrics endpoints are wired in `server/src/index.ts`. A narrative overview also exists in `docs/ARCHITECTURE.md`.

B Poker engine and cash game

B.1 What was programmed

Authoritative table logic lives in `server/src/logic/GameTable.ts` (seats, streets, pot and side pots, action legality, showdown ordering) and `server/src/logic/CashGameController.ts` (chip debits/credits, transitions between phases). Hand evaluation and category ordering use `server/src/logic/Evaluator.ts`. Real-time delivery and timers (e.g. `TURN_TIMER` with 20s auto-fold/check) are implemented in `server/src/sockets/game.gateway.ts` together with anti-cheat hooks. The client implements presentation and input in `client/src/pages/Game.tsx` and reusable components under `client/src/components/` (`PokerTable`, `ActionButtons`, `CommunityCards`, `chat`, `HUD` hooks). **All wagering state changes are validated and applied on the server**; the client never “decides” the winner.

B.2 Rules enforced

Texas Hold'em flow from blinds through preflop, flop, turn, river, and showdown; fold, check, call, raise semantics; minimum raise increments; side pots when players are all-in for different amounts. Waiting-room creation and join flows are handled by dedicated routes (`server/src/routes/waitingRoom.routes.ts`, `server/src/routes/game.api.routes.ts`) coordinated with the gateway.

B.3 Further reading

Structured logic notes: `Docs/detail_logique/backend/01_poker_moteur.md`, `Docs/detail_logique/backend/02_cashgame_controller.md`.

C Bot AI and practice integration

C.1 Programmed pipeline

Figures 1 and 2 in the main report depict the optional PyTorch policy ($36 \rightarrow 96 \rightarrow 64 \rightarrow 32 \rightarrow 4$, **196** post-input units, **11 972** trainable scalars in the linear stack) and the Node/Python pipeline. This appendix keeps the algebraic detail for heuristics and oracle mode. `POST /api/bot/action` (`server/src/routes/bot.routes.ts`) validates difficulty and cards, then calls `server/src/logic/botAI.ts`. Practice tables chain bot turns through `server/src/poker/services/practiceBotTurns.service.ts`; HTTP orchestration and optional expert delegation sit in `server/src/services/botAi.service.ts`. Four difficulties are supported: `easy`, `medium`, `hard`, `expert`.

C.2 Strength signal and decisions

Let H be the scalar hand score returned by `getHandValue` for the bot’s visible cards (two to seven cards). The code normalizes

$$\sigma = \min\left(1, \max(0, H/S_{\max})\right), \quad S_{\max} = 9 \cdot 15^5 + 14 \cdot 15^4,$$

matching the encoding used by `server/src/logic/Evaluator.ts`. Each difficulty combines σ with pot odds (`callAmount`, `potSize`), heads-up vs. multiway flags, and pseudo-random draws to return `FOLD`, `CALL`, `CHECK`, or `RAISE` with an optional raise amount (chips rounded via `intChips`). The easy tier, for example, injects naive bluffs on a free street with fixed probability and biases calls when facing small bets with medium strength.

C.3 Dispatch surface

`decideBotAction` in `server/src/logic/botAI.ts` switches on difficulty and routes to `easyBotDecision`, `mediumBotDecision`, `hardBotDecision`, or `expertBotDecision`. The HTTP layer (`server/src/services/botAi.service.ts`) may intercept `expert` to call an external model first.

C.4 Expert tier: optional Python service

When `AI_SERVICE_URL` is configured, `botAi.service.ts` issues `POST .../predict/poker` with a JSON payload built by `buildAiPayload` (compact card encoding, street, stacks, pot, position, optional opponent holes for oracle debugging). The JSON body is validated through `aiResponseSchema` (Zod): allowed actions include `ALL_IN`, which is lowered to a legal `RAISE/CALL/FOLD` via `convertAllIn`. Timeouts use `AbortController` and `env.aiServiceTimeoutMs`; structured logs record `expert_ai_decision` with latency and style. Any failure reverts to `expertBotDecision`.

C.5 Expert oracle path (TypeScript, no network)

For revealed opponent holes, `expertOracleDecision` computes hero strength $\sigma_{\text{hero}} = \text{normalizedHandStrength}$ and opponent strengths σ_i for each known hole pair. Sort $\sigma_{(1)} \geq \dots \geq \sigma_{(n)}$. For $n \leq 2$ let $S_{\text{opp}} = \sigma_{(1)}$. For $n \geq 3$ the code uses `compositeOpponentNormalizedStrength`:

$$S_{\text{opp}} = w \sigma_{(1)} + (1 - w) \sigma_{(\lfloor n/2 \rfloor)}, \quad w = \min(0.68, \max(0.35, 0.56 - 0.05(n - 3))),$$

so the policy blends the strongest visible villain with a median-like opponent instead of assuming everyone holds the nuts. The edge $\Delta = \sigma_{\text{hero}} - S_{\text{opp}}$ is compared to pot odds $\text{callAmount}/(\text{potSize} + \text{callAmount})$ and stack pressure $\text{callAmount}/\text{playerChips}$ to emit stylised lines (`value`, `bluff`, `pot_control`, `discipline`, etc.).

C.6 Further reading

Design notes: `Docs/IA_EXPERT_BOT.md`.

D Probabilities, Quantum HUD, and Monte Carlo helper

D.1 Division of responsibilities

The **Quantum HUD** (`client/src/components/QuantumHUD.tsx`, context `client/src/contexts/QuantumHUDContext.tsx`) displays *estimates* for the authenticated player. These are computed client-side for responsiveness and are **not** used to settle real-money outcomes (the product uses virtual chips). Server-side hidden-bet pricing uses separate tabular services (Appendix E).

D.2 Monte Carlo equity estimator

Implementation file: `client/src/utls/pokerMonteCarloEquity.ts`. Let N be the number of iterations (default $N = 2000$, constant `MONTE_CARLO_DEFAULT_ITERATIONS`). For each iteration the code:

1. Builds the remaining deck excluding known hero and community cards (`buildDeckExcluding`).
2. Draws, without replacement, the missing board cards and $2 \times n_{\text{opp}}$ opponent hole cards using a partial Fisher–Yates swap sampler (`drawWithoutReplacement`).
3. Evaluates the hero and each opponent seven-card hand with `evaluateSevenEval` (same taxonomy as the server evaluator).
4. Splits the simulated pot equally among all players that achieve the best score.

If $S_t \in \{0, 1/k\}$ is the hero’s share on iteration t when k players tie for the best score, the reported equity is

$$\hat{e} = \frac{1}{N} \sum_{t=1}^N S_t.$$

A histogram of category indices (0–9) is accumulated in parallel for UX badges.

D.3 Monte Carlo error budget (HUD analogue to RTP)

The HUD does **not** implement a house-banked game: there is no monetary **RTP** in the casino sense. The quantity $\hat{e}$ is an **equity** estimate (expected share of the pot at showdown under uniform completion of unknown cards). If e denotes the true value of that expectation under the model, then $\mathbb{E}[\hat{e}] = e$ (each draw is independent and fair with respect to the remaining deck). Because $0 \leq S_t \leq 1$,

$$\text{Var}(S_t) \leq \frac{1}{4}, \quad \text{Var}(\hat{e}) = \frac{\text{Var}(S_t)}{N}, \quad \text{SE}(\hat{e}) \leq \frac{1}{2\sqrt{N}}.$$

With the shipped default $N = 2000$, $\text{SE}(\hat{e}) \leq 1/(2\sqrt{2000}) \approx 0.01118$ (about 1.1 equity points at most in the worst-case variance bound).

D.4 Quantum Combi—exact enumeration formula

The promotional “Quantum Combi” category (documented for players in `Docs/final_report/QUANTUM_COMBI.adoc`) requires ranks $\{3, 5, 6, 7, 10\}$ each to appear **at least once** among the seven cards (two private + five community). All $\binom{52}{7}$ seven-card subsets are equally likely:

$$\binom{52}{7} = 133784560.$$

For each rank r in that set, let A_r be the event “no card of rank r among the seven”; then $|A_r| = \binom{48}{7}$. By the principle of inclusion–exclusion over five ranks,

$$\left| \bigcup_r A_r \right| = \sum_{k=1}^5 (-1)^{k+1} \binom{5}{k} \binom{52-4k}{7} = 133002736.$$

Hence the favourable count is $133784560 - 133002736 = 781824$ and

$$P(\text{Quantum Combi}) = \frac{781824}{133784560} \approx 0.00584 \approx 0.584\%.$$

D.5 Quantum Combi in the hand-ranking ladder (server and HUD)

Both `server/src/logic/Evaluator.ts` and `client/src/utils/pokerHandCategory.ts` use the same **best-of-seven** cascade. Categories are integers 0 (high card) through 9 (straight flush); a larger packed score beats a smaller one. After ruling out straight flush (9), four-of-a-kind (8), and full house (7), the code tests **Quantum Combi before flush or straight**. Quantum applies iff every rank in $\{3, 5, 6, 7, 10\}$ appears at least once in the seven rank values (Ace is coded as 14). Let V be the multiset of those seven values; subtract one occurrence of each mandatory rank; sort the remaining values descending and take the top two as kickers (κ_1, κ_2) . The score is

$$\text{packScore}(6, \kappa_1, \kappa_2) = 6 \cdot 15^5 + \kappa_1 \cdot 15^4 + \kappa_2 \cdot 15^3,$$

padding unused kicker slots with zeros in the implementation. Only if Quantum fails does the evaluator continue to flush (5), straight (4), three-of-a-kind, two pair, one pair, high card. Thus Quantum is positioned **strictly between full house and flush**, as a design choice documented in `Docs/final_report/QUANTUM_COMBI.adoc`.

E Hidden betting system

E.1 Programmed architecture

Hidden side markets are implemented under `server/src/poker/hiddenBets/`: typed markets and snapshots (`markets.ts`, `types.ts`), pricing tables (`pricing/pricingTables.ts`, `pricingTables.v1.ts`), live heuristics (`pricing/livePricing.ts`), quote and placement services, ledger context, and street-aware resolvers (`resolver/hiddenBetResolver.ts`). HTTP routes are mounted in `server/src/routes/hiddenBets.routes.ts`.

E.2 Odds and tables (example)

Live markets use tabulated decimal odds keyed by table parameters. For instance, `oddsPlayerWinsCurrentHand` maps the number of active players $n \in [2, 9]$ to a fixed grid (e.g. $n = 2 \rightarrow 1.9$, $n = 5 \rightarrow 3.0$, $n = 9 \rightarrow 4.6$); companion functions cover showdown vs. fold endings. These values are **heuristic** prices distanced from the pre-hand model (see code comments `hidden-bets-live-v1`), not Monte Carlo outputs from the HUD.

E.3 Further reading

`Docs/intern/HIDDEN_BETS_SPEC.md`, `Docs/detail_logique/backend/07_hidden_bets_moteur.md`.

F Tournaments and winner side bets

F.1 Runtime architecture (layers)

1. **API and validation** — `server/src/routes/tournament.routes.ts` (and related mounts in `server/src/index.ts`) expose creation, join, start, and spectator flows. Input normalisation lives in `tournament.create.validation.ts` (player counts, blind structure, schedule). Private tournaments hash join codes with bcrypt (`tournament.service.ts`).
2. **Persistence** — Prisma models `Tournament`, `TournamentPlayer`, `TournamentRound`, `TournamentTable`, etc., store lifecycle, elimination order, and links to underlying `gameId` values that reuse the cash-game engine.
3. **Scheduler** — `tournament.scheduler.ts` registers a timer every $\Delta T = 10\text{s}$ (via `setInterval`). On each tick it loads tournaments with `status = REGISTRATION_OPEN` and `startAt <= now` (bounded batch) and calls `startTournamentFromDb` with the Socket.IO server instance obtained from `app.get('io')`.
4. **Runtime orchestration** — `tournament.runtime.service.ts` coordinates bracket materialisation: it calls `buildOpeningRound / buildRoundFromSurvivors`, creates concrete `GameTable` instances through `tournamentTableFactory.ts`, emits `TOURNAMENT_TABLE_ASSIGNED` to per-user rooms, records eliminations when bust-out events arrive (`recordTournamentEliminationsIf`) and advances the state machine (`pending_other_tables`, `next_round_spawned`, `tournament_completed`).
5. **Winner side bets** — `winnerBets/tournamentWinnerBet.service.ts` handles ticket placement, settlement with tournament cancellation, and coordination with the wallet ledger.
6. **Recovery** — `tournament.recovery.service.ts` reconciles partially written bracket rows after crashes (invoked from server bootstrap).
7. **Gateway hooks** — `tournament.gatewayHook.ts` bridges table-level socket events into tournament bookkeeping.
8. **Client** — `client/src/features/tournament/` implements waiting room, ZipRush, in-tournament HUD, results, and winner-bet UI; routes under `/tournaments/...`

F.2 Bracket geometry and table sizing

Let m be the number of active players in a round ($m \geq 2$). `computeTableSizesForRound` in `server/src/tournament/bracket/tableSizes.ts` returns an integer partition $[s_1, \dots, s_k]$ such that $\sum_i s_i = m$, each $2 \leq s_i \leq 9$, using fixed templates for $m \leq 9$ (e.g. $m = 4 \rightarrow [2, 2]$, $m = 7 \rightarrow [4, 3]$) and, for $m \geq 10$,

$$k = \left\lceil \frac{m}{9} \right\rceil, \quad s_i \in \left\{ \left\lfloor \frac{m}{k} \right\rfloor, \left\lceil \frac{m}{k} \right\rceil \right\}$$

with the remainder distributed one seat at a time (see implementation). `assertTableSizesValid` enforces $\sum s_i = m$ and per-table bounds.

F.3 Seeded shuffle (deterministic bracket RNG)

`buildOpeningRound` and `buildRoundFromSurvivors` in `TournamentBracketBuilder.ts` take a 32-bit integer `seed`. They derive a PRNG `mulberry32(seed)` producing $u_n \in (0, 1)$ and apply Fisher–Yates shuffling so that the same `(playerIds, seed)` always yields the same table assignment (reproducible tests). The generator follows the published `mulberry32` recurrence (32-bit integer state, `Math.imul` mixes, bitwise rotations) as in the source file; the returned uniform is $u = ((t \oplus (t \gg 14)) \gg 0) / 2^{32}$.

F.4 Further reading

`Docs/intern/TOURNAMENT_FLOW_REAL_VS_EXPECTED.md`.

G Blackjack solo and multiplayer

G.1 Solo blackjack engine

Authoritative rules are implemented in `server/src/logic/blackjack.ts`. The server uses a six-deck shoe (`DECKS = 6`), shuffled in-place with Fisher–Yates (`shuffleShoe`). Card values follow standard rules: faces and tens count as 10, aces count as 11 when possible without exceeding 21, otherwise as 1. The dealer draws until the hard/soft total is at least 17; because aces are valued optimistically, the dealer **stands on soft 17** (e.g. A+6). Natural blackjack pays a 3:2 bonus on the stake (implemented as integer chip rounding per `intChips`); other wins pay 1:1; pushes return the stake. Double down is permitted only with exactly two player cards and issues one extra card.

G.2 HTTP and multi-table surfaces

Solo routes: `server/src/routes/blackjack.routes.ts`. Multiplayer rooms and seats: `server/src/routes/blackjackMulti.routes.ts`, controllers under `server/src/logic/BlackjackTableC`
`ts`, in-memory runtime maps with recovery under `server/src/blackjack/`. UI: `client/src/pages/Blackjack.tsx`, lobby/table flows, components in `client/src/components/blackjack/`.

G.3 Further reading

`Docs/detail_logique/backend/03_blackjack_solo_moteur.md`, `Docs/detail_logique/backend/04_blackjack_multi_moteur.md`.

H Slot machine, roulette, and casino wallet

H.1 Three-reel slot model

The server module `server/src/logic/slotMachine.ts` draws three independent reels. Symbol weights are

$$w_{\text{cherry}} = 32, w_{\text{lemon}} = 24, w_{\text{bell}} = 18, w_{\text{seven}} = 14, w_{\text{diamond}} = 8, \quad W = \sum_i w_i = 96.$$

One reel satisfies $\Pr[R=i] = w_i/W$. Triples satisfy $\Pr(R_1=R_2=R_3=s) = (w_s/W)^3$. Triple s pays total return multiplier $m_s \in \{5, 8, 10, 15, 20\}$ on the stake; an **exact pair** (two equal symbols, third different) pays multiplier 1 (stake returned, zero net profit); otherwise the payout multiplier is 0. RNG is injectable (`RandomIntFn`, default `crypto.randomInt` via `server/src/rng/rng.service.ts`).

H.2 Complete slot RTP (closed form, current constants)

Let $X = \text{winAmount}/\text{bet}$ be the **gross return multiplier** (total coins returned by the machine, including the stake on a push). Write R_1, R_2, R_3 for the three reel symbols.

Triple (three of a kind).

$$\Pr(\text{triple of } s) = \left(\frac{w_s}{W}\right)^3, \quad \Pr(\text{any triple}) = \sum_s \left(\frac{w_s}{W}\right)^3 = \frac{\sum_s w_s^3}{W^3}.$$

Numerically $\sum_s w_s^3 = 32768 + 13824 + 5832 + 2744 + 512 = 55680$ and $W^3 = 96^3 = 884736$, hence

$$\Pr(\text{any triple}) = \frac{55680}{884736} \approx 0.0629 \text{ (6.29\%)}.$$

Exact pair (not a triple). For a doubled symbol A ,

$$\Pr(\text{pair with } A \text{ doubled}) = \frac{3 w_A^2 (W - w_A)}{W^3}, \quad \Pr(\text{any pair}) = \sum_A \frac{3 w_A^2 (W - w_A)}{W^3}.$$

Numerically $\sum_A 3 w_A^2 (W - w_A) = 461952$, so $\Pr(\text{pair}) = 461952/884736 \approx 0.5221$ (52.21%).

Lose.

$$\Pr(\text{lose}) = 1 - \Pr(\text{triple}) - \Pr(\text{pair}) \approx 0.4149 \text{ (41.49\%)}.$$

Expected multiplier and RTP. Because outcomes partition $\{\text{lose, pair, triples}\}$,

$$\mathbb{E}[X] = \sum_s \left(\frac{w_s}{W}\right)^3 m_s + \Pr(\text{pair}) \cdot 1 = \frac{\sum_s m_s w_s^3 + 461952}{884736}.$$

Here $\sum_s m_s w_s^3 = 5 \cdot 32768 + 8 \cdot 13824 + 10 \cdot 5832 + 15 \cdot 2744 + 20 \cdot 512 = 384152$, so

$$\mathbb{E}[X] = \frac{384152 + 461952}{884736} = \frac{846104}{884736} \approx 0.9563.$$

The **return-to-player (RTP)** expressed as a percentage of the stake is $100 \mathbb{E}[X] \approx 95.63\%$. The **house edge** on the stake is $1 - \mathbb{E}[X] \approx 4.37\%$. The expected **net** gain per spin is $\mathbb{E}[G] = \mathbb{E}[\text{winAmount} - \text{bet}] = \text{bet} (\mathbb{E}[X] - 1) \approx -0.0437 \times \text{bet}$.

Variance (optional diagnostic). With the same X , the AsciiDoc reference reports $\text{Var}(X) \approx 3.12$ (standard deviation ≈ 1.77); hence $\text{Var}(G) = \text{bet}^2 \text{Var}(X)$ for a flat bet. See `Docs/SLOT_MACHINE.adoc` for the full derivation.

H.3 European roulette model

The server module `server/src/logic/roulette.ts` draws a single winning integer uniformly from $\{0, \dots, 36\}$ (`crypto.randomInt` through the same RNG service). Payouts follow the European **total-return** multipliers on the stake for winning pockets (straight 36, split 18, street 12, corner 9, six-line 6, dozen/column 3, even-money 2). Zero causes outside bets to lose unless covered.

H.4 Complete roulette RTP (all proportional inside bets)

Consider any bet line that covers exactly k distinct winning pockets ($1 \leq k \leq 18$ on the proportional ladder used in code) with European gross multiplier $m = 36/k$ applied to the stake when one of those pockets hits. Because each pocket has probability $1/37$,

$$\mathbb{E}\left[\frac{\text{payout}}{\text{bet}}\right] = \frac{k}{37} \cdot m = \frac{k}{37} \cdot \frac{36}{k} = \frac{36}{37} \approx 0.97297.$$

Hence every such line has the same **RTP** $= 36/37 \approx 97.297\%$ of the stake returned as gross payout in expectation, and the same **house edge** $= 1/37 \approx 2.703\%$ on the stake. The expected **net** gain per unit staked on a single line is

$$\frac{k}{37} \cdot m - 1 = -\frac{1}{37}.$$

When several independent bet lines are placed on one spin, expectation adds across lines; the server enforces caps (`ROULETTE_MAX_BET_CAP`, `ROULETTE_MAX_TOTAL_STAKE`, `ROULETTE_MAX_BETS_PER_SPIN`) before settlement. Wheel animation order is `EUROPEAN_WHEEL_ORDER`. Full payout tables: `Docs/final_report/ROULETTE.adoc`.

H.5 Wallet and ledger

Casino flows integrate with wallet/ledger helpers under `server/src/casino/` and `server/src/wallet/`. Additional narrative: `Docs/detail_logique/backend/05_roulette_moteur.md`, `Docs/detail_logique/backend/06_slot_moteur.md`, `Docs/final_report/ROULETTE.adoc`.

I Social graph, invitations, and peer loans

I.1 Programmed surfaces

Friends UI: `client/src/pages/Friends.tsx`. REST: `server/src/routes/friends.routes.ts`; poker invitations `server/src/routes/invitation.routes.ts` (also under `/api/invitations`); blackjack room invitations share patterns with the lobby. Peer loans: `server/src/routes/friendLoan.routes.ts`, business logic in `server/src/services/friendLoan.service.ts`, real-time updates via `server/src/services/friendLoan.emit.ts` and socket listeners on the client.

I.2 Further reading

`Docs/detail_logique/backend/09_friend_loans_moteur.md`.

J Progression, challenges, and leaderboards

J.1 Programmed logic

Shared rules in `server/src/logic/gamification.ts` (XP curves, badge triggers). Daily challenges and login streak are isolated modules (`server/src/dailyChallenges/`, `server/src/dailyLogin/`) with dedicated routers. Public leaderboards: `server/src/routes/leaderboard.routes.ts`; client pages under `/leaderboard` and profile summaries.

J.2 Further reading

`Docs/detail_logique/backend/10_gamification_moteur.md`.

K Authentication, profiles, and i18n

K.1 Programmed security and profiles

JWT issuance and verification on `server/src/routes/auth.routes.ts`; password hashing (bcrypt) and validation schemas (Zod). Optional TOTP 2FA: `server/src/routes/twofa.routes.ts`. Client auth flows: `client/src/pages/Auth.tsx`. Profile and avatar uploads: `client/src/pages/EditProfile.tsx` with binary fields or URLs on the Prisma `User` model. Internationalization bundles live in `client/src/i18n/locales/` (five languages); guest-only screens force English overlays where documented in Section 11.

L Operations, Redis, CI, and packaging

L.1 Programmed operations

Redis clients for Socket.IO horizontal scale-out: `server/src/config/socketIoRedis.ts`. Rate limiting and action logs share Redis primitives configured in `server/src/index.ts`. Observability helpers: `server/src/observability/`. CI pipeline: `.gitlab-ci.yml`. Admin web console: `client/src/pages/AdminConsole.tsx` with `server/src/routes/adminConsole.routes.ts`. Dangerous admin/dev routes are compiled out or gated when `env.isProduction` is true.

L.2 Kaniko container builds (GitLab CI)

The `docker-build` stage invokes **Kaniko** executor `v1.23.2-debug` with an empty container `entrypoint` so the shell can call `/kaniko/executor`. **(1)** `docker:build—server/` `Dockerfile`, context `server/`, destinations `${CI_REGISTRY_IMAGE}:${CI_COMMIT_SHORT_SHA}` and `:latest`; runs on `develop`, `main`, and `release/*`. **(2)** `docker:build:client—client/` `Dockerfile`, build-arg `VITE_API_URL` (override in GitLab CI/CD variables for behind-nginx paths), `KUBERNETES_MEMORY_REQUEST/LIMIT` raised to reduce OOM during Kaniko layer snapshots; destinations `:client-${CI_COMMIT_SHORT_SHA}` and `:client-latest`; same branch rules as (1). **(3)** `docker:build:ai-service—context server/ai-service/`, FastAPI stack for the optional poker prediction endpoint; destinations `:ai-${CI_COMMIT_SHORT_SHA}` and `:ai-latest`; **only** `release/*` (commented in YAML to spare runner disk on `develop/main`). All three jobs use `-compressed-caching=false` for predictable memory on small Kubernetes runners. `client/Dockerfile` documents Kaniko-oriented Dockerfile layering (grouped RUN steps) to avoid oversized intermediate snapshots after `npm ci`.

L.3 Electron and Capacitor build matrix

The npm scripts are declared in `client/package.json` (see build block for electron-builder). `npm run build:electron` runs Vite in electron mode; `electron:build:win`, `electron:build:mac`, `electron:build:linux`, and `electron:build:all` invoke `electron-builder` with the corresponding platform flags. Windows uses NSIS (`.exe` installer); macOS uses the default mac target (typically `.dmg` plus update artefacts); Linux emits `AppImage` and `.deb` as declared under `build.linux.target`. `publish:win` / `publish:mac` add `-publish` always for CI/CD uploads to the generic update server configured in `build.publish`.

Capacitor: `npm run build:cap` then `npx @capacitor/cli sync` copies web assets into `android/` and `ios/` native shells (`@capacitor/android`, `@capacitor/ios` 8.3.0). `npm run cap:android` / `cap:ios` open the native IDEs; `cap:run:*` supports live-reload against a LAN API host for device testing.

M Test coverage metrics and definitions

M.1 Jest ratios

Let L be the set of executable lines inside `collectCoverageFrom` in `server/jest.config.cjs`. For each metric $M \in \{\text{lines}, \text{statements}, \text{functions}, \text{branches}\}$,

$$\text{cov}_M = \frac{\# \text{ elements of } M \text{ hit during tests}}{\# \text{ elements of } M \text{ instrumented}}.$$

CI enforces $\text{cov}_{\text{lines}} \geq 0.80$, $\text{cov}_{\text{statements}} \geq 0.80$, $\text{cov}_{\text{functions}} \geq 0.80$, and $\text{cov}_{\text{branches}} \geq 0.65$. A May 2026 local run reported $(\text{cov}_{\text{lines}}, \text{cov}_{\text{statements}}, \text{cov}_{\text{functions}}, \text{cov}_{\text{branches}}) = (86.56\%, 83.69\%, 88.75\%, 67.90\%)$ on that scope. Heavy files such as `server/src/logic/botAI.ts` and large controllers are deliberately excluded from the denominator so the metric tracks testable business modules (see comments in the Jest config).

M.2 Vitest ratios (client)

Vitest with Istanbul uses the same hit/total idea over `client/src/**/*.ts,tsx` minus tests (see `client/vite.config.ts`). Because most screens depend on sockets and browser APIs, the **global** line coverage stays low unless many components receive dedicated tests; quality is also asserted through Playwright flows and manual regression lists.

M.3 Relation to tournaments

Bracket helpers (`tableSizes.ts`, `TournamentBracketBuilder.ts`) sit inside the Jest coverage glob; their golden tests contribute directly to the aggregate in Table 3. See Appendix F for the mathematics they implement.

M.4 Further reading

`Docs/test_coverage.md`, `Docs/TESTING_GUIDE.md`, `Docs/RAPPORT_TESTS.md`.

N External references and compendium

- Initial formal requirements PDF: Docs/Cahier_de_charge_6A_PI.pdf (not embedded here).
- Auto-generated wide-coverage technical compendium: Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md (use as index; verify claims against code).
- Security dossiers: Docs/Architecture/SECURITE_A_TO_Z.md, Docs/security/.
- HTTP and socket maps: Docs/detail_logique/network/01_http_api_map.md, Docs/detail_logique/network/02_socket_events_map.md.
- Combination analysis report: Docs/final_report/QUANTUM_COMBI.adoc.
- Slot RTP and outcome algebra (AsciiDoc): Docs/SLOT_MACHINE.adoc.
- Roulette payouts and expectation: Docs/final_report/ROULETTE.adoc.

References

- [1] Meta. *React documentation* (React 19). <https://react.dev/> (accessed May 2026).
- [2] Microsoft. *TypeScript handbook*. <https://www.typescriptlang.org/docs/> (accessed May 2026).
- [3] Vite contributors. *Vite guide*. <https://vite.dev/guide/> (accessed May 2026).
- [4] OpenJS Foundation. *Node.js runtime documentation*. <https://nodejs.org/docs/latest/api/> (accessed May 2026).
- [5] OpenJS Foundation. *Express web framework*. <https://expressjs.com/> (accessed May 2026).
- [6] Socket.IO project. *Socket.IO server and client (v4)*. <https://socket.io/docs/v4/> (accessed May 2026).
- [7] Prisma Data Inc. *Prisma ORM documentation*. <https://www.prisma.io/docs> (accessed May 2026).
- [8] PostgreSQL Global Development Group. *PostgreSQL documentation*. <https://www.postgresql.org/docs/> (accessed May 2026).
- [9] Redis Ltd. *Redis documentation*. <https://redis.io/docs/> (accessed May 2026).
- [10] Redux team. *Redux Toolkit and RTK Query*. <https://redux-toolkit.js.org/> (accessed May 2026).
- [11] i18next authors. *i18next documentation*. <https://www.i18next.com/> (accessed May 2026).
- [12] Zod maintainers. *Zod schema validation*. <https://zod.dev/> (accessed May 2026).
- [13] OpenJS Foundation. *Jest testing framework*. <https://jestjs.io/> (accessed May 2026).
- [14] Vitest contributors. *Vitest*. <https://vitest.dev/> (accessed May 2026).
- [15] Microsoft. *Playwright end-to-end testing*. <https://playwright.dev/> (accessed May 2026).
- [16] Tailwind Labs. *Tailwind CSS documentation*. <https://tailwindcss.com/docs> (accessed May 2026).
- [17] Schwaber, K., Sutherland, J. *The Scrum Guide* (2020). <https://scrumguides.org/scrum-guide.html> (accessed May 2026).
- [18] GitLab Inc. *GitLab CI/CD documentation*. <https://docs.gitlab.com/ee/ci/> (accessed May 2026).
- [19] Université de Strasbourg. *Faculté des mathématiques et informatique*. <https://mathinfo.unistra.fr/> (accessed May 2026).